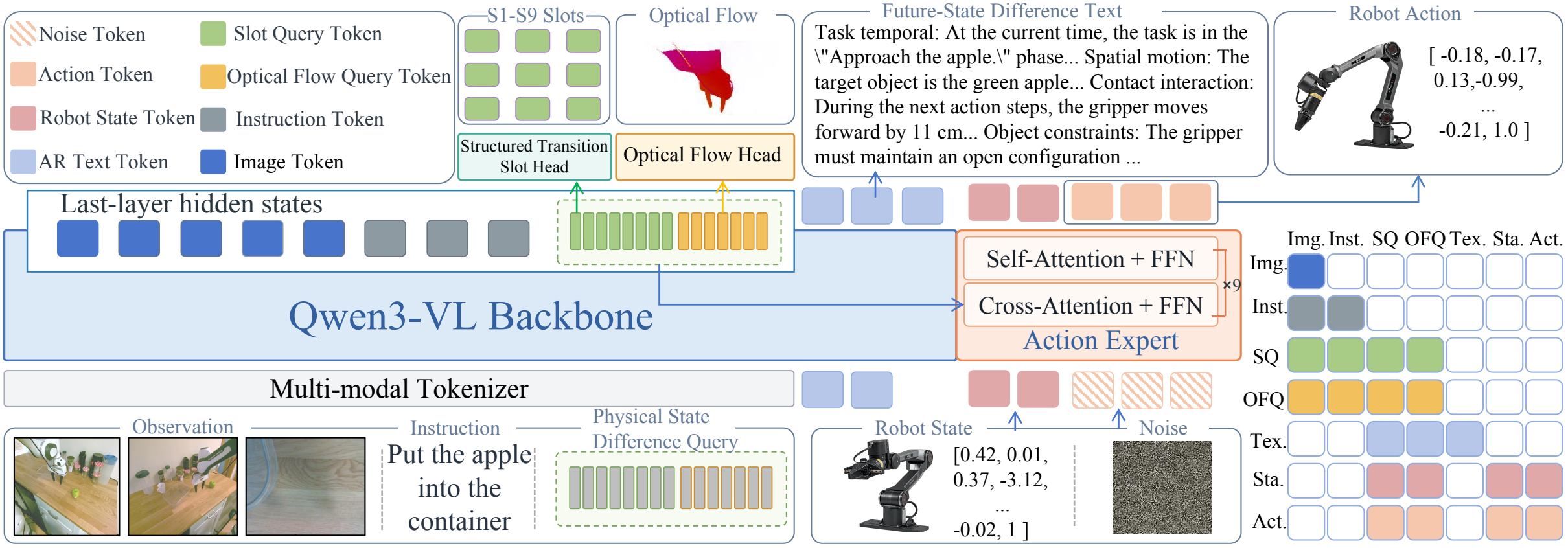

Noise Token
Slot Query Token
Action Token
Optical Flow Query Token
Robot State Token
Instruction Token
AR Text Token
Image Token
S1-S9 Slots
Optical Flow
Structured Transition Slot Head
Optical Flow Head
Future-State Difference Text
Task temporal: At the current time, the task is in the \"Approach the apple.\" phase... Spatial motion: The target object is the green apple... Contact interaction: During the next action steps, the gripper moves forward by 11 cm... Object constraints: The gripper must maintain an open configuration ...
Robot Action
[-0.18, -0.17, 0.13,-0.99, ... -0.21, 1.0]
Last-layer hidden states
Qwen3-VL Backbone
Self-Attention + FFN
Cross-Attention + FFN
×9
Action Expert
Img. Inst. SQ OFQ Tex. Sta. Act.
Multi-modal Tokenizer
Observation
Instruction
Put the apple into the container
Physical State Difference Query
Robot State
[0.42, 0.01, 0.37, -3.12, ... -0.02, 1]
Noise